



SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	04/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	04/07/2026

Assigned user to Sign Off	Oraib AlKheetanRojas
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	

Operation Assessed	RB-CAR Autonomous Buggy – Operation
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Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

The RB-CAR is a 4 × 4 electric research vehicle derived from a Polaris Ranger EV and fitted with servo-actuated steering, traction drive and onboard computer for manual or autonomous driving. Routine work includes charging, pre-drive checks, tele-operation, autonomous navigation and data collection within the lab or on adjacent outdoor test pads. Activities involve engaging high-capacity lead-acid batteries, operating safety PLCs, using emergency-stop hardware, and manoeuvring the buggy at speeds up to 25 km/h.

Reason for Risk Assessment

To document and control any health, safety and equipment-damage risks introduced by active use of the RB-CAR in research and demonstration tasks.

Overall Current Risk

High Risk

   

Average Number of Persons Affected

2  Employees

2  Students

Current Rating	Hazard Information	Task Status
High Risk	Vehicles The buggy can reach 25 km/h and weighs ~ 800 kg; loss of control, sharp turns or excessive speed may cause collision or rollover, endangering occupants and bystanders.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Key-switch and mode selector force neutral on start-up. - Dash-mounted and remote E-stop buttons latch spring-loaded handbrake and cut traction drive. - Safety PLC monitors laser scanner, brake switch and remote link; movement inhibited on fault. - Seat-belt use, speed-mode limiter (Low/Max Range/High) and AWD lock restrict torque on test beds. - Operator training and restricted-access test zone with floor markings.	

Potential Rating	Additional Controls Required
High Risk	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk	Moving Parts Rotating wheels, steering linkages and the automated handbrake present crush and entanglement points during driving or maintenance.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Guarded under-body and steering rack; service panels secured. - Controller disables steering motor when key is off. - Only trained personnel may work on suspension or drive components with main power connector unplugged. - Clear signage and exclusion zone during autonomous runs.	

Potential Rating	Additional Controls Required
Moderate Risk	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk	Electricity Eight series-parallel lead-acid batteries supply 48 V DC at high current; damaged cables, improper charging or wet conditions could cause electric shock or short circuit.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Batteries wired as SELV with colour-coded, insulated connectors; main power connector isolates entire pack. - On-board Delta-Q charger interlocked to charge-status indicator and requires vented hood open. - Periodic water-level checks, torque to 8 ft lb on terminals, and visual inspection before use. - No jumper-cable starts; auxiliary loads limited to fused 12 V/24 V DC-DC rails.	

Potential Rating	Additional Controls Required
Moderate Risk	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	 Fire Over-charging, internal battery fault or cable short may ignite hydrogen gas or lead-acid vapours; hot motor/controller can also kindle leaking electrolyte.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Charger has temperature and voltage cut-outs and cycles to trickle mode automatically. - Charging only in ventilated area (= 5 air changes h⁻¹) with hood and doors open; “no-cover-while-charging” rule. - Lab equipped with Class C dry-powder extinguishers and staff trained in their use. - Periodic inspection of insulation, cables and DC bus fuse (25 A).	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	 Hot surfaces Traction motor, controller heat-sink and brake discs exceed 80 °C after prolonged driving; accidental contact can burn skin or ignite debris.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Warning decal at rear hatch; components shielded within sealed cabinet. - Five-minute cool-down idle enforced in SOP before servicing. - Gloves required for any work inside rear electronics bay; key off and batteries isolated first.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.